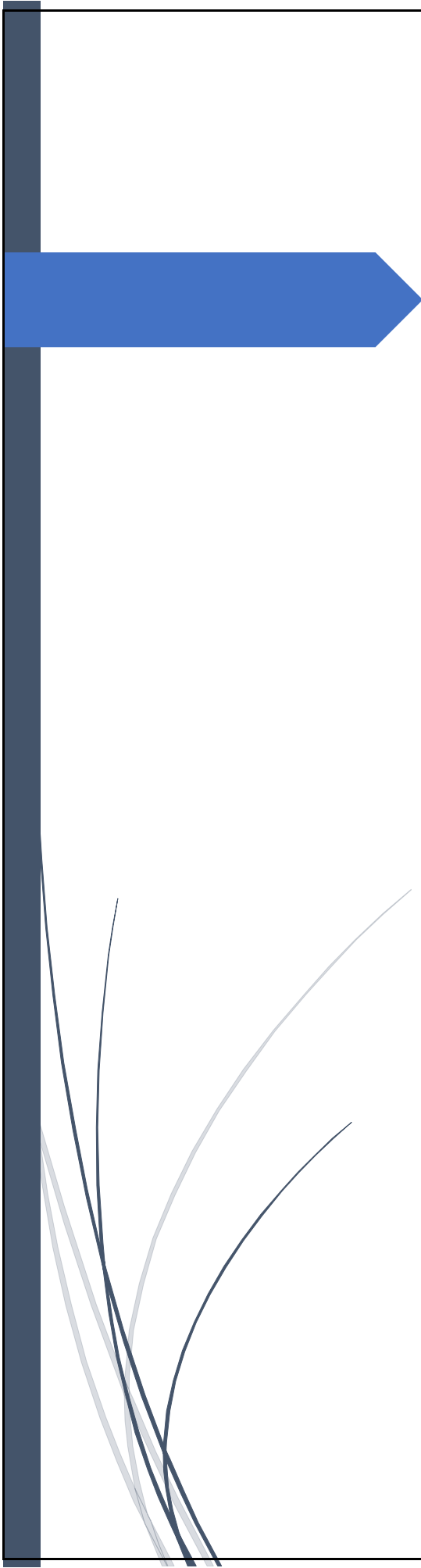




Terms of Reference

Student Administration System (SAS)
for Nadi Muslim College

Group Number: 14

Semester: 2, 2025

CS241 Software Design & Implementation

Group Member Details

Name	Student ID
Ranveer Singh	S11230141
Mohammed Afeef	S11229568
Mohammed Suhail	S11230995
Ronil Prasad	S11231541
Shivan Prasad	S11231502

Background

Most secondary schools in Fiji including, Nadi Muslim College still rely on the paper-based systems to manage their student records, attendance and grades. This often results in problems such as lost files, difficulties in tracking student progress, and weak communication among teachers and parents. Instead of focusing fully on teaching, Staff end up spending too much time dealing with paperwork. To address this issue our dedicated team is working on developing a Student Administration System (SAS) that is specifically designed for Nadi Muslim College, with aim of making these processes more efficient and reliable.

Purpose of the Team

Our team's main goal is to build a student administration system that makes managing student information much easier for schools. We will be using Java and Spring boot to design a platform that will be simple for teachers, parents and students to use. The system is intended to provide solutions to real-life challenges the school faces on a daily basis, that would ensure that its not just functional but also user-friendly.

Available Resources

To support and complete this project, we have access to university computer labs equipped with development tools, high speed internet, licensed software such IntelliJ IDE and MySQL Workbench as well as library resources and printing facilities. In addition, each team member has a personal laptop with the necessary Java Setup, home internet access, and a wide range of online tutorials and learning resources to support ongoing development.

Skill Inventory

Member	Role	Technical Skills	Analysis & Design Skills	Research, Writing & Presentation Skills	Managerial Skills
Ranveer Singh	Project Manager & Backend Developer	• Java programming (CS214) • Spring Boot framework	• Requirements engineering (CS230) • Data structures (CS112) • System architecture	• Technical documentation (UU200) • Project reporting • Team communication	• Project coordination • Task delegation
Mohammed Afeef	System Designer	• Java programming (CS214) • Database systems (IS222)	• Requirements analysis (CS230) • Use case modelling • Algorithm design (CS112)	• Design specifications • Technical writing (UU200) • Design presentations	• Design coordination • Cross-functional collaboration
Mohammed Suhail	Database Designer	• Database design (IS222) • Java connectivity (CS214) • SQL optimization	• Database normalization (IS222) • Data modeling • Data structures (CS112)	• Database documentation • Technical specifications (UU200) • Data analysis reporting	• Timeline management • Backend coordination
Ronil Prasad	Frontend Developer & Business Analyst	• Figma, Adobe XD • HTML/CSS (IS221) • Java basics (CS214)	• UI/UX design principles • Requirements gathering (CS230) • User-centered analysis	• User research interviews • Design presentations	• Task prioritization • User feedback management
Shivan Prasad	Quality Tester & Security Developer	• Java security (CS214) • Testing frameworks	• Testing methodology • Security requirements (CS230) • Performance analysis (CS112)	• Test documentation • Quality reports • Technical writing (UU200)	• Testing coordination • Quality oversight

Roles

Member	Job	Description
Ranveer Singh	Project Manager & Backend Developer	Monitor the rest of the project members work, use Spring Boot to build the main system as well as manage the database connection.
Mohammed Afeef	System Designer	Design the whole system, create use cases and write implementation plan.
Mohammed Suhail	Database Designer	Design the database and data models and make sure that every record is stored properly and securely.
Ronil Prasad	Frontend Developer & Business Analyst Design	Design and build user interface, communicate with users and stakeholders to understand their requirements and create a user-friendly user interface.
Shivan Prasad	Quality Tester & Security Developer	Test the whole system for errors and make sure it works correctly, handle user logins and permissions.

Decision Making Policy

As a group, we discussed the system requirements and component before making all the significant decisions. Ranveer, the project manager will have the authority to decide whether we need to decide right away or we're deadlocked.

Code of Conduct and Ethical Issue

Every participant is required to actively participate in all the sessions and show up for planned meetings for development. For those who are not able to attend must notify the project manager in advance and turn in their contributions.

Every piece of code needs to adhere to Spring-Boot practices and Java coding standards. Each piece of code needs to undergo peer review before they are added to the main repository. Extra review cycles, code revision, or team sessions would be essential for violations.

Every development project needs to be unique and given proper credits for any outside libraries or framework. User privacy and data security must be taken into account regarding design choices. All the code violations must be immediately replaced, or discussed with the course coordinator.

Weekdays should include productive and professional team communications. While all the technical inquiries must be answered within 24 hours. The main emphasis for code review should be technical merit and areas for improvement. Private discussions and team mediation are to be used to resolve issues when necessary.

Team Participation Strategies

To discuss progress and coordinate code, we converse twice a week. For challenging features, members pair up, alternating partners every week to exchange knowledge. Viber is utilized for daily progress as well so that we can discuss and have quick questions.

Every code is checked by peers prior to being merged with the master project. We employ Git, whose branches are properly organized and whose commit messages are clean. Weekly tests ensure everything runs smoothly.

Viber is our primary communication for day-to-day communications as well as technical queries. All project files are shared via Google Drive.

In development, junior members are assisted by senior members. Learning Java technologies and latest Spring Boot features is done at a predetermined time. Individual R&D work on individual system components.

Complex Computing Problem Demonstration

We are designing a multi-user solution, which will allow multiple users to be logged in simultaneously without issues. As instructors enter marks, the system automatically derives grades and fills in attendance data.

Only the authorized people will have access to sensitive data, such as grades and personal data, because proper user authentication will be implemented for security. Automatic chart and report generation, which reflects students' progress over time, is also provided by the system.

Peer Review Strategies

These include early code reviews in which assigned reviewers examine every member's work; cross-functional reviews where frontend developers examine backend code and vice versa; integration reviews where the entire team examines the functionality of the integrated system; and end-of-project quality checks against project requirements and coding standards.

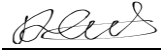
Comprehensive coverage of mandatory features and functionality, correctness of technical implementation, compliance with Java and Spring Boot best practices, maintainability and quality of code and documentation, security features, and error handling are the top review criteria.

Feedback integration emphasizes heavily on agreement within teams on final implementation decisions, several review loops for difficult features, logging of all review comments and resolution mechanisms, and feedback that is directed towards improving code.

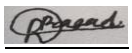
By signing below, each team member agrees to the following statements:

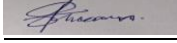
- a) I participated in formulating the standards, roles, and procedures as stated in this contract.
- b) I understand that I am obligated to abide by these terms and conditions.
- c) I understand that if I do not abide by these terms and conditions, I will suffer the consequences as stated in this contract.

Ranveer Singh (S11230141)  **Date:** 21/08/25

Mohammed Afeef (S11229568)  **Date:** 21/08/25

Mohammed Suhail (S11230995)  **Date:** 21/08/25

Ronil Prasad (S11231541)  **Date:** 21/08/25

Shivan Prasad (S11231502)  **Date:** 21/08/25

Date Created: 21/08/2025

Final Submission: 31 August 2025

This Terms of Reference/Team Contract governs our CS241 project collaboration and may be updated by unanimous team consent as development requirements evolve.